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Note from author

These notes are mostly based on the syllabus and past papers

Fair warning: they might look short (2–3 pages), but I've condensed everything down to keep the important stuff while removing the extra yap. That said, I wouldn't recommend using these as quick revision notes, but think of them as detailed notes covering almost everything

Spotted a mistake? Suggestions?

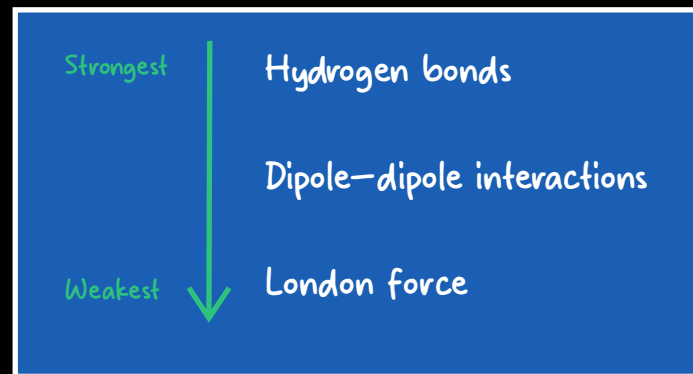
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Intermolecular forces

Introduction

Intermolecular forces are the attractive forces **between molecules**. They are much **weaker than** the bonds within a **molecule** (covalent bonds), but they are responsible for **determining** a substance's **physical properties** like melting point, boiling point, viscosity, and surface tension. They are also referred to as **non-bonded** interactions hence why they mainly affect physical properties not chemical identity.



London Forces (LDFs)

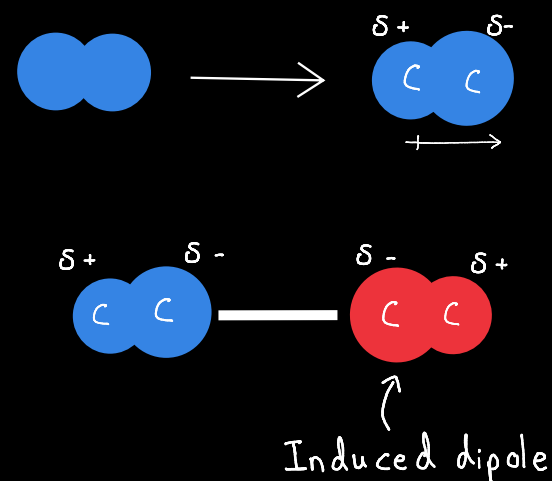
How it works

Due to the **constant and random motion** of electrons (giving them a probabilistic location), there will be **instants when more electrons accumulate on one side of the atom than the other**.

creating a **temporary, instantaneous dipole**: one side has a slight excess of negative charge (δ^-), the other side a slight electron deficit, making it partially positive (δ^+). This is not a permanent polarity.

Its dipoles **induces a dipole** on that **neighboring atom**, and is aligned opposite to the first dipole (δ^+ near δ^- , and δ^- near δ^+).

These two synchronized, opposite temporary charges now experience an **electrostatic attraction**. This attraction is the **London force**.

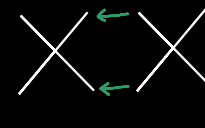
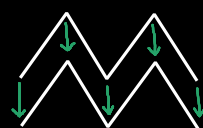


Main Factors that affect size of London Forces

- **Number of electrons**: **More electrons** → more distance between electrons and nucleus (less attraction) → **greater fluctuations** in electron density → **larger instantaneous and induced dipoles created** making London forces **stronger** so more energy is required to overcome them (boiling points will increase)
- **Shape & Size of molecule**: **More points of contact** (greater contact surface) → **Greater overall London force** as instantaneous dipole-induced dipole forces exist at each point of contact between the molecules. Hence **straight chains** are **stronger** than **branched chains** (fewer the points of contact) and **longer chains** are **stronger** than **shorter chains**.

A weaker intermolecular forces mean that substance has higher volatility.

E.g. Alkanes become more volatile if the branching increases (Contact surface), while it becomes less volatile as chain grows (more electrons + more contact points)



Pentane is a straight chain with a greater contact surface (less compact) so it has greater London forces both have same electron number



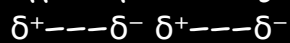
Pentane is greater number of electrons (greater molecular mass) so the London forces are stronger

Permanent Dipole-Dipole interactions

How it works

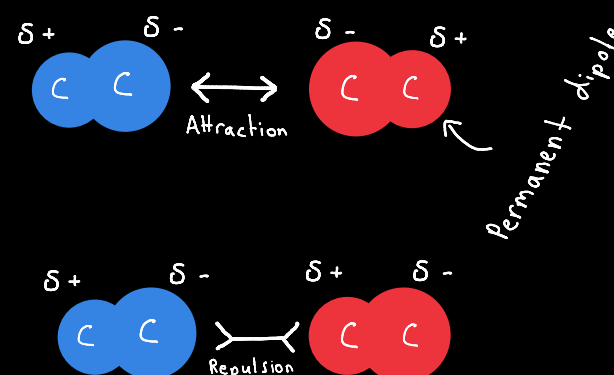
Polar molecules have a **permanent dipole** due to **difference in electronegativity** (e.g., HCl, H₂O, NH₃)

Opposite partial charges align:



This creates a net electrostatic attraction stronger than London forces alone, but they could also be aligned oppositely (meaning they repel)

It is also possible that a permanent dipole induces a dipole in a nearby molecule



Hydrogen bonding

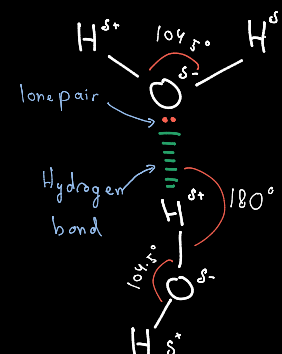
A special, strong type of intermolecular or sometimes intramolecular (though not required) attraction between:

- A **hydrogen** atom that is covalently **bonded to a highly electronegative atom** (N, O, or F).
- A **lone pair** of electrons on another **highly electronegative atom** (N, O, or F) in a different

molecule (or a different part of the same molecule).

The "Donor" and "Acceptor" Concept

- **Hydrogen Donor:** The molecule (or group) that **supplies the hydrogen atom (H)** covalently bonded to N, O, or F.
- **Electron Pair Acceptor (Acceptor):** The **atom with a lone pair** (on N, O, or F) that attracts the partially positive hydrogen.



Factors affecting strength:

- The electronegativity: $\text{H-F} > \text{H-O} > \text{H-N} > \text{H-S}$

- Nature of the Acceptor: A hydrogen bond to an ion is stronger than that to a neutral molecule

- The average number of hydrogen atoms per molecule: even though $\text{H-F} > \text{H-O}$, the hydrogen bonding in H_2O is stronger than HF as it has an average of 2 hydrogen bonds per molecule while HF has 1

Why N, O, and F?

- **High Electronegativity:** They pull bonding electrons away from H very strongly.
- **Small Atomic Size:** This allows the exposed hydrogen nucleus (a bare proton) to get very close to the lone pair on the acceptor atom, creating a strong electrostatic attraction.
- **Result:** The H atom has a very high partial positive charge (δ^+)—it is essentially "deshielded." Though it is no confined to these atoms

Nature of the Interaction: Beyond a Simple Force

A hydrogen bond is not merely a weak intermolecular attraction; it exhibits clear evidence of partial bond formation, placing it in a distinct category **stronger than London dispersion or dipole-dipole forces**. (shown by the fact that is directional with an angle of 180°)

Answering method for question comparing compounds boiling points (strength):

1. Identify forces/interaction (e.g. London, dipole-dipole, hydrogen bonding) for each one

- **London dispersion forces (LDFs)** — present in all molecules; strength increases with more electrons/molar mass/size.
- **Permanent dipole-dipole interactions** — present in polar molecules (due to difference in electronegativity).
- **Hydrogen bonding** — present if H is bonded to F, O, or N

2. Compare electronegativity (No always required)

3. FOR the same interactions (e.g. only London /dipole dipole)

—mention the factor causing the difference (e.g. more electron/ greater difference in electronegativity (or mention if they have the same strength))

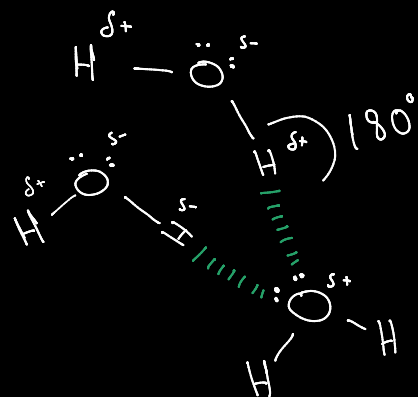
4. FOR different interaction (only found in one)

—State which one is stronger (H-bond > dipole-dipole > LDFs)

5. "More energy is required to break {Compound}"

H₂O

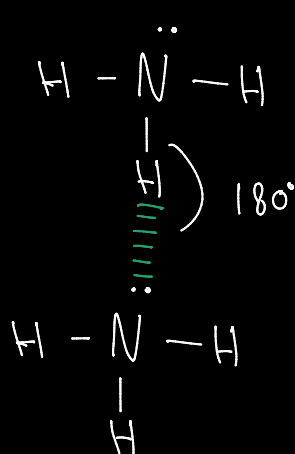
2 lone pairs (acceptor), 2 hydrogens (donators)



Us

NH₃

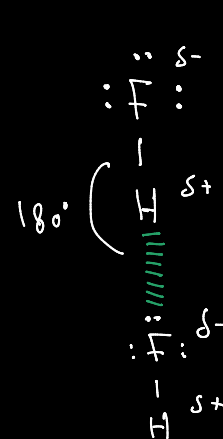
1 lone pair (acceptor), 3 hydrogens (donor)



Us

HF

3 lone pairs (acceptor), 1 hydrogen (donor)



1. They all have similar London forces strength as they have the same electron number, and all 3 also have hydrogen bonds.
2. F is **more electronegative** so its **hydrogen bonds are stronger than that of H₂O and NH₃** (HF is more polar)
3. However, H₂O forms **twice as much hydrogen bonds** (2 on average) compared to HF and NH₃ which form 1 on average.

So H₂O > HF > NH₃

Hydrogen bonding

Anomalous properties of water (due to hydrogen bonding)

• Its high melting and boiling temperature compared to similar molecule

Without H-bonds, boiling points would increase down Group 16 (H_2S , H_2Se , H_2Te) due to London forces, however, water's strong and extensive H-bond network gives it extra cohesive energy → needs more energy to break the bonds → Higher boiling/melting points (it can form up to 4 hydrogen bonds)

• Density of Ice < Density of Water

Open hexagonal lattice in ice:

Due Each to lattice / ring structure in ice

H-bonds are longer than covalent bonds

Creating a more open, low-density structure with large empty spaces.

Upon melting:

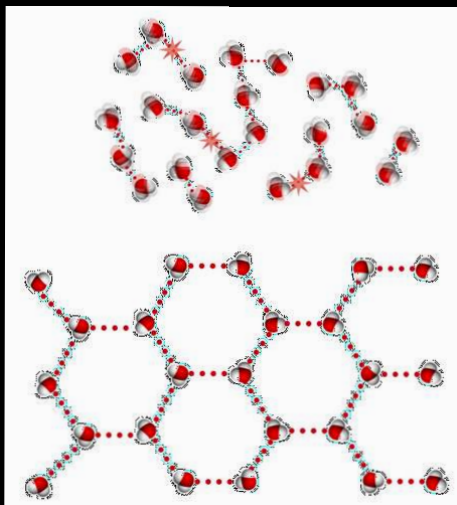
Some H-bonds break.

Molecules pack more closely in the liquid.

Result: Liquid water is denser than ice → ice floats.

Water (liquid)

Ice (Solid)



Alcohols Vs. Alkanes

Alcohols have one $-\text{OH}$ group attached. But that one $-\text{OH}$ group makes a HUGE difference because it can form hydrogen bonds. Alkanes only have London dispersion forces (weak, temporary attractions).

For Alkanes:

they only have London dispersion forces. These forces get stronger as the molecule gets bigger because bigger molecules have more electrons and more surface area for temporary dipoles to form.

Note that branching the alkane weakens the LDFs as it reduces points of contact

For Alcohols:

They have

1. Hydrogen bonding from the $-\text{OH}$ group (STRONG)
2. Dipole-dipole interactions
3. London dispersion forces from the carbon chain

Think of it like this: Alcohols = alkanes + (OH) hydrogen bond

HOWEVER as their chains grow the the relative strength of H-bonds decreases as LDFs are the only forces increasing (while there is only 1 OH (polar bond) for all alcohols) → they begin behaving more similarly (LDFs is now the predominant force)

The strength on intermolecular interactions can be determined using the **enthalpy of vaporization**, which is the amount of energy required to convert a (mol) liquid into a gas at a given pressure and temperature.

A greater enthalpy → stronger the interactions holding them together.

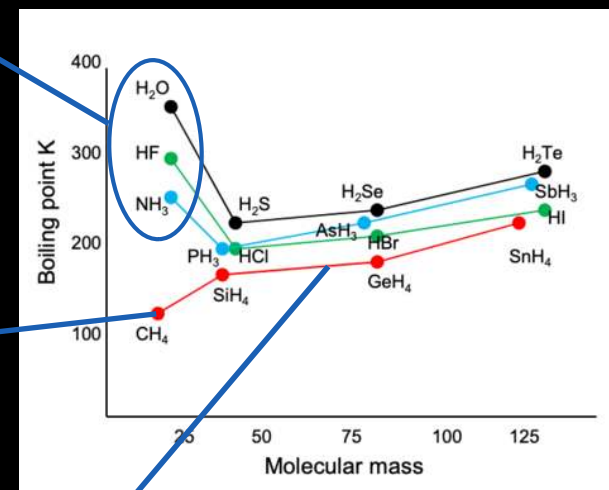
A lower enthalpy → weaker the interactions holding them together.

the trends in boiling temperature down the group

H_2O , HF , NH_3 have much higher boiling points

Reason: They have strong H-bonding + London forces.

CH_4 has no H-bond
it follows normal trend



General increase down the group after (H_2S , PH_3 , HCL)

Reason: they have no H-bond only LDFs and dipole-dipole interactions. Down the group chain grows → LDFs increases

Solubility

Basic principle

For a substance to dissolve,

- The solute particles must be separated from each other and then become surrounded by solvent particles
- The force of attraction between the solute and solvent particles must be strong enough to overcome the solvent-solvent (e.g. pure water) forces and the solute-solute (e.g. pure water) forces

Solubility of Covalent Molecules

a) Non-polar molecules in non-polar solvents (e.g. hexane in benzene)

Both hexane and benzene are non-polar and have only London (dispersion) force.

The London forces between hexane and benzene molecules are similar in strength to those in the pure liquids, so enough energy is released to break bond between the pure liquids. (it dissolves)

b) Non-polar molecules in water (e.g. hexane in water)

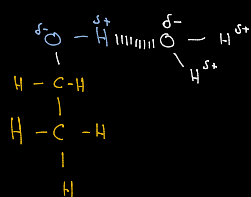
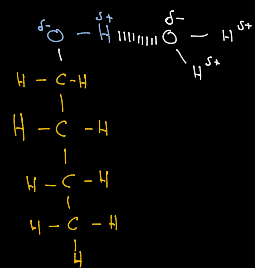
Water molecules form a strong hydrogen-bonds with each other. Non-polar molecules (e.g. iodine, hexane) cannot form hydrogen bonds — only weak London forces. The hydrogen bonds between water molecules >>> London forces between water and non-polar molecules. (Breaking water's hydrogen bonds requires significant energy, which is not compensated by forming weak solute-solvent interactions.) (it does not dissolve)

c) Polar molecules in polar solvents (e.g. Ethanol in water)

Ethanol has an -OH group that can form hydrogen bonds, and Water also forms hydrogen bonds.

So Ethanol can form H-bonds with water of similar strength to the original water-water and ethanol-ethanol hydrogen bonds, making it energetically feasible. (it dissolves)

Trend for Alcohols: Solubility in water decreases with increasing hydrocarbon chain length because the non-polar alkyl chain (interacting only via weak London dispersion forces) becomes dominant. The single -OH group cannot compensate for the disruption to water's hydrogen-bond network. (London forces become predominant)



Section that can only form London forces

Section that can form hydrogen bonds

This section grows in size as chain length increases

This section is the same (length) for all regardless of size

Polar Bonds That Do NOT Form Hydrogen Bonds

A molecule can have polar bonds but still be incapable of forming hydrogen bonds if it lacks a hydrogen atom bonded directly to F, O, or N (or lacks a lone pair on a highly electronegative atom to accept hydrogen bonds).
e.g. Carbonyl Compounds, and Halogenoalkanes

Dissolving ionic compound in water

Step 1: Water Approaches the Ionic Lattice

Water molecules are polar: δ^- on oxygen, δ^+ on hydrogens

They orient themselves around ions:

Oxygen (δ^-) faces cations (positive ions)

Hydrogens (δ^+) face anions (negative ions)

Step 2: Water molecules break ionic lattice

Water molecules form ion-dipole bonds with surface ions

These attractions compete with the ionic bonds in the lattice

With enough water molecules, the electrostatic attraction between ions is weakened

Individual ions break away from the crystal surface (detached)

Step 3: Hydration Shell Forms (Exothermic)

Each ion becomes surrounded by water molecules

Hydration shell forms via strong ion-dipole interactions

Energy is released during this step (hydration energy)

Hydration is the process by which the water molecules arrange around the ions and it releases energy called the hydration energy.

The hydration energy is used to compensate for energy used for breaking lattice

IF:

Hydration energy $\geq$ lattice energy $\rightarrow$ Soluble

IF

Hydration energy $<$ lattice energy $\rightarrow$ Insoluble

note: if Hydration energy released is slightly less than Lattice energy required the substance dissolves but solution becomes colder

